

```
Last login: Tue Jun 30 16:12:51 on ttys000
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch-200mb-2\
>
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch-200mb-2
psql (18.4)
Type "help" for help.
```

```
tpch-200mb-2=# SET ROLE alice;
```

```
EXPLAIN (ANALYZE, VERBOSE, BUFFERS)
```

```
select
    s_name,
    s_address
from
    supplier_secure,
    nation
where
    s_suppkey in (
        select
            ps_suppkey
        from
            partsupp_secure
        where
            ps_partkey in (
                select
                    p_partkey
                from
                    part
                where
                    p_name like '%ivory%'
            )
        and ps_availqty > (
            select
                0.5 * sum(l_quantity)
            from
                lineitem_secure
            where
                l_partkey = ps_partkey
                and l_suppkey = ps_suppkey
                and l_shipdate >= date '1995-01-01'
                and l_shipdate < date '1995-01-01' + interval '1' year
            )
        )
    and s_nationkey = n_nationkey
    and n_name = 'FRANCE'
order by
    s_name;
```

```
SET
```

```
WARNING: Log 1: Planner Input:
```

```
DETAIL: {QUERY
```

```
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
  :alias <>
  :eref
  {ALIAS
  :aliasname supplier_secure
  :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
  }
}
:rtkind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
  :alias
  {ALIAS
  :aliasname s
  :colnames <>
  }
  :eref
  {ALIAS
  :aliasname s
  :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_ph
one" "s_acctbal" "s_comment")
  }
}
:rtkind 0
:relid 18153
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rtperminfos (
  {RTPERMISSIONINFO
  :relid 18153
  :inh true
  :requiredPerms 2
  :checkAsUser 10
  :selectedCols (b 8 9 10 11 12 13 14)
  :insertedCols (b)
  :updatedCols (b)
  }
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
  :rtindex 1
  }
)
:quals
{BOOLEXP
:boolop and
:args (
  {OPEXP
  :opno 1756
  :opfuncid 1720
  :opresulttype 16
  :opretset false
  }
  }
}
```

```

:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 6
    :vartype 1700
    :vartypmod 983046
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
  }
  {FUNCEXP
    :funcid 1740
    :funcresulttype 1700
    :funcresultset false
    :funcvariadic false
    :funcformat 2
    :funccollid 0
    :inputcollid 0
    :args (
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 0 0 0 0 0 0 0 ]
      }
    )
    :location -1
  }
)
:location -1
)
{SUBLINK
:sublinktype 0
:sublinkid 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname n
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname n
      :colnames ("n_nationkey" "n_name" "n_regionkey" "n_
comment")
    }
    :rtekind 0
    :relid 18130
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname r
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname r
      :colnames ("r_regionkey" "r_name" "r_comment")
    }
    :rtekind 0
    :relid 18136
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 18130
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 10)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 18136
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 9)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
:fromList (
  {RANGETBREF
    :rtindex 1
  }
  {RANGETBREF
    :rtindex 2
  }
)
:quals
{BOOLEXP
:boolop and
:args (
  {OPEXP
    :opno 96

```

```

:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
  }
  {VAR
    :varno 1
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location -1
  }
)
:location -1
)
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  }
  {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
  }
)
:location -1
)
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 2
    :varattno 2
    :vartype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 2
    :location -1
  }
)
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
  }
  :resno 1
  :resname ?column?
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
)
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>

```

```

        :limitOption 0
        :rowMarks <>
        :setOperations <>
        :constraintDeps <>
        :withCheckOptions <>
        :stmt_location -1
        :stmt_len -1
    }
    :location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname s_supkey
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname s_name
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1043
:vartypmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname s_address
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname s_nationkey
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname s_phone
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname s_acctbal
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 6

```

```

:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:varatype 1043
:varitypmod 105
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname s_comment
:ressortgroupref 0
:resorigtbl 10153
:resorigcol 7
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier true
:relid 18621
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 18130
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18621
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18130
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
}
:quals
{BOOLEXPR
:boolop and
:args (
{SUBLINK
:subLinkType 2
:subLinkId 0
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varatype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 136
}
{PARAM
:paramkind 2
:paramid 1
:paramtype 23
:paramitypmod -1
:paramcollid 0
:location -1
}
)
}
:location 146
}
)
:operName ("=")
:subselect

```

```

{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTRY
  :alias <>
  :eref
    {ALIAS
    :aliasname partsupp_secure
    :colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_su
    pplycost" "ps_comment")
    }
  }
:rtekind 1
:subquery
  {QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs false
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSubLinks true
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity false
  :hasGroupRTE false
  :isReturn false
  :cteList <>
  :rtable (
    {RANGETBLENTRY
    :alias
      {ALIAS
      :aliasname ps
      :colnames <>
      }
    :eref
      {ALIAS
      :aliasname ps
      :colnames ("ps_partkey" "ps_suppkey" "ps_availqty"
      "ps_supplycost" "ps_comment")
      }
    }
    :rtekind 0
    :relid 18163
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :infromCl true
    :securityQuals <>
    }
  )
:rteperminfos (
  {RTEPERMISSIONINFO
  :relid 18163
  :inh true
  :requiredPerms 2
  :checkAsUser 10
  :selectedCols (b 8 9 10 11 12)
  :insertedCols (b)
  :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
  :fromlist (
    {RANGETBLREF
    :rtindex 1
    }
  )
  :quals
    {BOOLEXP
    :boolop and
    :args (
      {OEXPR
      :opno 521
      :opfuncid 147
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location -1
        }
      {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 100 0 0 0 0 0 0 0 ]
      }
      )
      :location -1
      }
    {SUBLINK
    :sublinkType 0
    :sublinkId 0
    :testexpr <>
    :operName <>
    :subselect
      {QUERY
      :commandType 1
      :querySource 0
      :canSetTag true
      :utilityStmt <>
      :resultRelation 0
      :hasAggs false
      :hasWindowFuncs false
      :hasTargetSRFs false
      :hasSubLinks false
      :hasDistinctOn false
      :hasRecursive false
      :hasModifyingCTE false
      :hasForUpdate false

```

```

:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname s
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname s
      :colnames ("s_suppkey" "s_name" "s_addr
        ess" "s_nationkey" "s_phone" "s_acctbal
        " "s_comment")
    }
    :rtekind 0
    :relid 18153
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 18153
    :inh true
    :requiredPerms 2
    :checkAsUser 18
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
}
:quals
{BOOLEXP
  :boolop and
  :args (
    {OPEXP
      :opno 96
      :opfuncid 65
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 1
          :location -1
        }
        {VAR
          :varno 1
          :varattno 2
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 1
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 2
          :location -1
        }
      )
    }
    :location -1
  )
  {OPEXP
    :opno 1756
    :opfuncid 1720
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 6
        :vartype 1700
        :vartypmod 983046
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 6
        :location -1
      }
      {FUNCEXP
        :funcid 1740
        :funcresulttype 1700
        :funcset false
        :funcvariadic false
        :funcformat 2
        :funccollid 0
        :inputcollid 0
        :args (
          {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbval true
            :constisnull false
            :location -1
            :constvalue 4 [ 0 0 0 0 0 0 0
              0 ]
          }
        )
      }
    }
    :location -1
  )
}
:location -1
}
:location -1
}
:location -1
}
:location -1
}
:location -1
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23

```

```

        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
}
:resno 1
:resname ps_partkey
:ressortgroupref 0
:resorigtbl 18163
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
:resno 2
:resname ps_supkey
:ressortgroupref 0
:resorigtbl 18163
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
}
:resno 3
:resname ps_availability
:ressortgroupref 0
:resorigtbl 18163
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 1700
:vartypmod 983846
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
}
:resno 4
:resname ps_supplycost
:ressortgroupref 0
:resorigtbl 18163
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1043
:vartypmod 203
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1

```

```

        :varattnosyn 5
        :location -1
    }
    :resno 5
    :resname ps_comment
    :ressortgroupref 0
    :resorigtbl 18163
    :resorigcol 5
    :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier true
:relid 18625
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18625
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
):jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{SUBLINK
:subLinkType 2
:subLinkId 0
:testexpr
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:verno 1
:varattno 1
:varrtype 23
:varrtymod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 316
}
{PARAM
:paramkind 2
:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
}
)
:location 327
}
:operName ("=")
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname part
:colnames ("p_partkey" "p_name" "p_mfgr" "p_b
rand" "p_type" "p_size" "p_container" "p_reta
ilprice" "p_comment")
}
:rtkind 0
:relid 18141
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18141
:inh true

```

```

        :requiredPerms 2
        :checkAsUser 0
        :selectedCols (b 8 9)
        :insertedCols (b)
        :updatedCols (b)
    }
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{OPEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{RELABELTYPE
:arg
{VAR
:varno 1
:varattno 2
:vartype 1043
:vartypmod 59
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 581
}
}
:resulttype 25
:resulttypmod -1
:resultcollid 100
:relabelformat 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 593
:constvalue 11 [ 44 0 0 0 37 105 118 111 1
14 121 37 ]
}
}
:location 588
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 411
}
:resno 1
:resname p_partkey
:ressortgroupref 0
:resorigtbl 18141
:resorigcol 1
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 327
}
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{FUNCEXPR
:funcid 1740
:funcresulttype 1700
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 657
}
)
}
:location -1
}
{SUBLINK
:subLinkId 0
:subLinkType 4
:testexpr <>
:operName <>

```

```

:subselect
{
  QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs true
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSublinks false
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity false
  :hasGroupRTE false
  :isReturn false
  :ctelist <>
  :rtable (
    {RANGETBLENTRY
     :alias <>
     :eref
     {ALIAS
      :aliasname lineitem_secure
      :colnames ("l_orderkey" "l_partkey" "l_sup
pkey" "l_linenumber" "l_quantity" "l_exten
dedprice" "l_discount" "l_tax" "l_returnfl
ag" "l_linestatus" "l_shipdate" "l_commitd
ate" "l_receiptdate" "l_shipinstruct" "l_s
hipmode" "l_comment")
      }
    }
  :rtekind 1
  :subquery
  {
    QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSublinks true
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE false
    :isReturn false
    :ctelist <>
    :rtable (
      {RANGETBLENTRY
       :alias
       {ALIAS
        :aliasname l
        :colnames <>
        }
       :eref
       {ALIAS
        :aliasname l
        :colnames ("l_orderkey" "l_partkey"
"l_suppkey" "l_linenumber" "l_quantit
y" "l_extendedprice" "l_discount" "
l_tax" "l_returnflag" "l_linestatus"
"l_shipdate" "l_commitdate" "l_rece
iptdate" "l_shipinstruct" "l_shipmod
e" "l_comment")
        }
       :rtekind 0
       :relid 18194
       :inh true
       :relkind r
       :rellockmode 1
       :perminfoindex 1
       :tablesample <>
       :lateral false
       :inFromCl true
       :securityQuals <>
       }
    )
    :rteperminfos (
      {RTEPERMISSIONINFO
       :relid 18194
       :inh true
       :requiredPerms 2
       :checkAsUser 10
       :selectedCols (b 8 9 10 11 12 13 14 15
16 17 18 19 20 21 22 23)
       :insertedCols (b)
       :updatedCols (b)
       }
    )
  }
  :jointree
  {
    FROMEXPR
    :fromlist (
      {RANGETBLREF
       :rtindex 1
       }
    )
  }
  :quals
  {
    BOOLEXP
    :boolop and
    :args (
      {SUBLINK
       :subLinkType 0
       :subLinkId 0
       :testexpr <>
       :operName <>
       :subselect
       {
        QUERY
        :commandType 1
        :querySource 0
        :canSetTag true
        :utilityStmt <>
        :resultRelation 0
        :hasAggs false
        :hasWindowFuncs false
        :hasTargetSRFs false
        :hasSublinks false
        :hasDistinctOn false
        :hasRecursive false
        :hasModifyingCTE false
        :hasForUpdate false
        :hasRowSecurity false
        :hasGroupRTE false
        :isReturn false
        :ctelist <>
        :rtable (
          {RANGETBLENTRY
           :alias
           {ALIAS
            :aliasname o
            :colnames <>
            }
           :eref
           {ALIAS
            :aliasname o
            :colnames ("o_orderkey"
"o_custkey" "o_orderstat
us" "o_totalprice" "o_or
derdate" "o_orderpriorit
y" "o_clerk" "o_shipprio
rity" "o_comment")
            }
           :rtekind 0
           :relid 18182

```

```

:inh true
:relkind r
:rellockmode 1
:permindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18182
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b
)
:varlevelsup 0
:varreturningtype
0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b
)
:varlevelsup 1
:varreturningtype
0
:varnosyn 1
:varattnosyn 1
:location -1
}
}
:location -1
}
{SCALARARRAYOPEXP
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 1
:varattno 6
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b
)
:varlevelsup 0
:varreturningtype
0
:varnosyn 1
:varattnosyn 6
:location -1
}
{ARRAYEXP
:array_typeid 1014

:array_collid 100
:element_typeid 10
42
:elements (
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false

:location -1
:constvalue 12 [ 4
8 0 0 0 49 45 85 8
2 71 69 78 84 ]
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false

:location -1
:constvalue 10 [ 4
0 0 0 0 50 45 72 7
3 71 72 ]
}
)
:multidims false
:list_start -1
:list_end -1
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr

```

```

(CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0
0 0 0 0 ]
)
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
)
:override 0
:onConflict <>
:ReturningOldAlias <>
:ReturningNewAlias <>
:ReturningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
)
:location -1
)
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey"
"ps_suppkey" "ps_availqt
y" "ps_supplycost" "ps_c
omment")
}
:rtekind 0
:relid 18163
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rtepermissions (
{RTEPERMISSIONINFO
:relid 18163
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
}
:varlevelsup 0
:varreturningtype
0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
}
:varlevelsup 1
}
}

```

```

        :varreturningtype
        0
        :varnosyn 1
        :varattnosyn 2
        :location -1
    }
}
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 1
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b
        )
        :varlevelsup 0
        :varreturningtype
        0
        :varnosyn 1
        :varattnosyn 2
        :location -1
    }
    {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b
        )
        :varlevelsup 1
        :varreturningtype
        0
        :varnosyn 1
        :varattnosyn 3
        :location -1
    }
}
:location -1
}
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b
        )
        :varlevelsup 0
        :varreturningtype
        0
        :varnosyn 1
        :varattnosyn 3
        :location -1
    }
    {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 0
        0 0 0 0 0 0 ]
    }
}
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 4 [ 1 0 0 0
            0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>

```

```

:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname l_suppkey
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname l_linenummer
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
}

```

```

:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)

```

```

        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 14
        :location -1
    }
    :resno 14
    :resname l_shipinstruct
    :ressortgroupref 0
    :resorigtbl 18194
    :resorigcol 14
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 15
    :vartype 1042
    :vartypmod 14
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 15
    :location -1
    }
    :resno 15
    :resname l_shipmode
    :ressortgroupref 0
    :resorigtbl 18194
    :resorigcol 15
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 16
    :vartype 1043
    :vartypmod 48
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 16
    :location -1
    }
    :resno 16
    :resname l_comment
    :ressortgroupref 0
    :resorigtbl 18194
    :resorigcol 16
    :resjunk false
    }
    }
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDecls <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :security_barrier true
    :relid 18629
    :inh false
    :relkind v
    :rellockmode 1
    :perminfoindex 1
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    )
    :rteperminfos (
    {RTEPERMISSIONINFO
    :relid 18629
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 9 10 12 18)
    :insertedCols (b)
    :updatedCols (b)
    }
    )
    )
    :jointree
    {FROMEXPR
    :fromlist (
    {RANGETBLREF
    :rtindex 1
    }
    )
    }
    :quals
    {BOOLEXP
    :boolop and
    :args (
    {OPEXP
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location 945
    }
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 957
    }
    )
    )
    )
    )

```

```

:location 955
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 1012
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 1024
}
)
:location 1022
}
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 1079
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 1098
:constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 ]
}
)
:location 1090
}
{OPEXPR
:opno 2345
:opfuncid 2330
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 1155
}
{OPEXPR
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 1173
:constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 ]
}
{CONST
:consttype 1186
:consttypmod 327679
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 1197
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 12 0 0 0 ]
}
)
:location 1186
}
)
:location 1166
}
)
:location 1008
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{OPEXPR
:opno 1760
:opfuncid 1726
:opresulttype 1700
:opretset false
:opcollid 0
:inputcollid 0

```

```

:args (
  {CONST
    :consttype 1700
    :consttypmod -1
    :constcollid 0
    :constlen -1
    :constbyval false
    :constisnull false
    :location 752
    :constvalue 8 [ 32 0 0 0 -1 -128 -120 1
      9 ]
  }
  {AGGREF
    :aggfnoid 2114
    :aggtype 1700
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 0
    :aggargtypes (o 1700)
    :aggdirectargs <>
    :args (
      {TARGETENTRY
        :expr
        {VAR
          :varno 1
          :varattno 5
          :vartype 1700
          :vartypmod 983046
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 5
          :location 762
        }
        :resno 1
        :resname <>
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
      }
    )
    :aggorder <>
    :aggdistinct <>
    :aggfilter <>
    :aggstar false
    :aggvariadic false
    :aggkind n
    :aggresorted false
    :agglevelsup 0
    :aggsplit 0
    :aggno -1
    :aggtransno -1
    :location 758
  }
)
:location 756
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 671
}
)
:location 669
}
)
:location 653
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 2
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location 198
    }
    :resno 1
    :resname ps_suppkey
    :ressortgroupref 0
    :resorigtbl 18625
    :resorigcol 2
    :resjunk false
  }
)
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

```

:location 146
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 1254
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 1268
}
)
:location 1266
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location 1292
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 1301
:constvalue 10 [ 40 0 0 0 70 82 65 78 67 69 ]
}
)
:location 1299
}
)
:location 1250
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 51
}
:resno 1
:resname s_name
:ressortgroupref 1
:resorigtbl 18621
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1043
:vartypmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 67
}
:resno 2
:resname s_address
:ressortgroupref 0
:resorigtbl 18621
:resorigcol 3
:resjunk false
}
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>

```

```

:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: max_parallel_hazard_walker: Param paramid=1 in_safe_list=0 → reason=outer_reference
WARNING: max_parallel_hazard_walker: Param paramid=1 in_safe_list=0 → reason=outer_reference
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 propparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 propparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 propparallel=s (s=safe r=restricted u=unsafe)
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem_secure
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname l
:colnames <>
}
}
:eref
{ALIAS
:aliasname l
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
}
:rtekind 0
:relid 18194
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18194
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname o
:colnames <>
}
}
:eref
{ALIAS
:aliasname o

```

```

:colnames ("o_orderkey" "o_custkey" "o_orderstatus"
"o_totalprice" "o_orderdate" "o_orderpriority" "o_
clerk" "o_shippriority" "o_comment")
}
}
:rtekind 0
:relid 18182
:inh true
:relkind r
:rellockmode 1
:petminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18182
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromList (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
:location -1
}
)
{SCALARARRAYOEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:neqfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 1
:varattno 6
:varatype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
)
{ARRAYEXPR
:array_typeid 1014
:array_collid 100
:element_typeid 1042
:elements (
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 12 [ 48 0 0 0 49 45 85 82 71 6
9 78 84 ]
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 50 45 72 73 71 7
2 ]
}
)
:multidims false
:list_start -1
:list_end -1
:location -1
}
)
:location -1
}
)
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true

```

```

:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppkey" "ps_availability"
"ps_supplycost" "ps_comment")
}
:rtekind 0
:relid 18163
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18163
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
:location -1
}
)
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16

```

```

:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
  }
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  }
)
:location -1
}
}
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  }
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
  }
)
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname l_orderkey
    :ressortgroupref 0
    :resorigtbl 18194
    :resorigcol 1
    :resjunk false
  }
)

```

```

}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname 1_partkey
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname 1_supkey
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname 1_linenummer
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname 1_quantity
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname 1_discount
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
}

```

```

:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 1042
:varypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 1042
:varypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:varypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:varypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:varypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:varypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location -1
}
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 1042
:varypmod 14
:varcollid 100
:varnullingrels (b)

```

```

        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 15
        :location -1
    }
    :resno 15
    :resname l_shipmode
    :ressortgroupref 0
    :resorigtbl 18194
    :resorigcol 15
    :resjunk false
    }
    {TARGETENTRY
    :expr
        {VAR
        :varno 1
        :varattno 16
        :vartype 1043
        :vartypmod 48
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 16
        :location -1
        }
    :resno 16
    :resname l_comment
    :ressortgroupref 0
    :resorigtbl 18194
    :resorigcol 16
    :resjunk false
    }
    }
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :security_barrier true
    :relid 18629
    :inh false
    :relkind v
    :rellockmode 1
    :perminfoindex 1
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    }
    :rteperminfos (
    {RTEPERMISSIONINFO
    :relid 18629
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 9 10 12 18)
    :insertedCols (b)
    :updatedCols (b)
    }
    )
    :jointree
    {FROMEXPR
    :fromlist (
    {RANGETBLREF
    :rtindex 1
    }
    )
    :quals (
    {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location 945
    }
    {PARAM
    :paramkind 1
    :paramid 0
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location 957
    }
    )
    :location 955
    }
    {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location 1012
    }
    {PARAM
    :paramkind 1
    :paramid 1
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location 1024
    }
    )
    }
    }

```

```

    }
  }
  :location 1022
}
{OEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 1079
}
  {CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 1098
:constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 -1 ]
}
)
:location 1090
}
{OEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 1155
}
  {CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 -39 -79 49 -115 -1 -1 ]
}
)
:location 1166
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
:expr
  {OEXPR
:opno 1760
:opfuncid 1726
:opresulttype 1700
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location 752
:constvalue 8 [ 32 0 0 0 -1 -128 -120 19 ]
}
  {AGGREF
:aggfnoid 2114
:aggtype 1700
:aggcollid 0
:inputcollid 0
:aggtransno 0
:aggargtypes (o 1700)
:aggdirectargs <>
:args (
  {TARGETENTRY
:expr
  {VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 762
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar false
:aggvariadic false
:aggkind n
:aggresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 758
}
)
:location 756
}
:resno 1
:resname ?column?

```

```

:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1098 :opfuncid 1090 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 11 :vartype 1
1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 11 :location -1} {CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constby
val true :constisnull false :location -1 :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 -1 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof
false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_
serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_buck
etsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 2345 :opfuncid 2338 :opresulttype 16 :opretset false :opcollid 0 :inputco
llid 0 :args ({VAR :varno 1 :varattno 11 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 11 :location -1} {CONST :consttype 1114
:consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 32 -39 -79 49 -115 -1 -1 ]} :location -1} :is_pushed_down true :can_join false :pseudoconst
ant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :lef
t_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left fals
e :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opr
esulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varatt
nosyn 2 :location -1} {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false
:is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :right_reld
s (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 2 :
vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype
23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const
true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :l
eft_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23
:vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :l
ocation -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds
(b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :out
er_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnos
yn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkin
d 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left fa
lse :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})

WARNING: max_parallel_hazard_walker: Param paramid=0 in safe_list=0 * reason=outer reference

WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1098 :opfuncid 1090 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 11 :vartype
1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 11 :location -1} {CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constb
yval true :constisnull false :location -1 :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 -1 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproo
f false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :orclause <> :rinfo_
serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_buc
ketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 2345 :opfuncid 2338 :opresulttype 16 :opretset false :opcollid 0 :inputc
ollid 0 :args ({VAR :varno 1 :varattno 11 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 11 :location -1} {CONST :consttype 1114
:consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 32 -39 -79 49 -115 -1 -1 ]} :location -1} :is_pushed_down true :can_join false :pseudocons
tant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :lef
t_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left fal
se :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :op
resulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varat
tnosyn 2 :location -1} {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone fals
e :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b 1) :right_reld
s (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 2 :
vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype
23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const
true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :l
eft_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 2
3 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :
location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds
(b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :ou
ter_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varno
syn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkin
d 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left f
alse :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = false | reason = PROPARALLEL_REST
RICTED

WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:      QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTRE false
:isReturn false
:relList <>
:table (
  {RANGETBLENTRY
    :alias
      {ALIAS
        :aliasname 1
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname 1
        :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenummer" "l_quali
        ty" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines
        tatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "
        l_shipmode" "l_comment")
      }
    :rtekind 0
    :reldid 18194
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBLENTRY
    :alias
      {ALIAS
        :aliasname o
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname o
        :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o
        _orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
      }
    :rtekind 0
    :reldid 18182
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
  }
)
```

```

:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtckind 0
:relid 18163
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18194
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18182
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18163
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBREF
:rtindex 1
}
)
:quals (
{OPEXPR
:opno 1098
:opfuncid 1098
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 1098
:constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 -1 ]
}
)
:location 1098
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
{PARAM
:paramkind 1
:paramid 0
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location 957
}
)
:location 955
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (

```

```

        {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location -1
        }
        {PARAM
        :paramkind 1
        :paramid 1
        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location 1024
        }
    )
    :location 1022
}
)
}
:rarg
{RANGETBLREF
:rtindex 2
}
:usingClause <>
:join_using_alias <>
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
:location -1
}
{SCALARARRAYOEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 6
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
{CONST
:consttype 1014
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 48 [ -64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0
1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 7
3 71 72 0 0 ]
}
)
:location -1
}
)
:alias <>
:rtindex 0
}
:rarg
{RANGETBLREF
:rtindex 3
}
:usingClause <>
:join_using_alias <>
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1

```

```

        :varattnosyn 2
        :location -1
    }
}
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 3
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 2
        :location -1
    }
    {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location -1
    }
)
:location -1
}
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 3
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 3
        :location -1
    }
    {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
    }
)
:location -1
}
)
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull true
            :location -1
            :constvalue <>
        }
        :resno 1
        :resname l_orderkey
        :ressortgroupref 0
        :resorigtbl 18194
        :resorigcol 1
        :resjunk false
    }
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull true
            :location -1
            :constvalue <>
        }
        :resno 2
        :resname l_partkey
        :ressortgroupref 0
        :resorigtbl 18194
        :resorigcol 2
        :resjunk false
    }
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull true
            :location -1
            :constvalue <>
        }
        :resno 3
        :resname l_suppkey
        :ressortgroupref 0
        :resorigtbl 18194
        :resorigcol 3
        :resjunk false
    }
)
{TARGETENTRY
:expr
{CONST

```

```

:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
:resno 4
:resname l_linenummer
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1700
:consttypmod 983046
:constcollid 0
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1700
:consttypmod 983046
:constcollid 0
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
}
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1700
:consttypmod 983046
:constcollid 0
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 5
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 5
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
}

```

```

:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 29
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 14
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1043
:consttypmod 48
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 16
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRUCTINFO :clause {OEXPR :opno 1098 :opfuncid 1098 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varrattno 11 :vartype 1
082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 11 :location -1} {CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constby
val true :constisnull false :location -1 :constvalue 4 [-34 -8 -1 -1 -1 -1 -1]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof
false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_
serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_buck
etsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheoperator 0 :right_hasheoperator 0} {RESTRUCTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid
0 :args ({VAR :varno 1 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 2 :location -1} {PARAM :paramkind 1 :paramid 0
:paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :se
curity_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 9 :eval_cost.startu
p -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCMEMBER :em_expr {VAR :varno 1 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 2 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3)}} :ri
ght_em {EQUIVALENCMEMBER :em_expr {PARAM :paramkind 1 :paramid 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdoma
in {JOINDOMAIN :jd_relds (b 1 2 3)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheoperator 0 :right_hasheoperator 0}
{RESTRUCTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varrattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (
b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 3 :location -1} {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down tr
ue :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_re
lds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 11 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :mergeopfamilies (o 1976) :left_
em {EQUIVALENCMEMBER :em_expr {VAR :varno 1 :varrattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 3 :location -1} :em_relds
(b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3)}} :right_em {EQUIVALENCMEMBER :em_expr {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmo
d -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3)}} :outer_is_left false :hashjoinoperator 0 :left_bucket
size -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheoperator 0 :right_hasheoperator 0})
size -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheoperator 0 :right_hasheoperator 0})
max_parallel_hazard_walker: Param paramid=0 -> reason=outer_reference

```



```

{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname s
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname s
      :colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_ph
one" "s_acctbal" "s_comment")
    }
    :rtekind 0
    :relid 18153
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 18153
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 0 9 10 11 12 13 14)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
    :rtindex 1
  }
)
:quals
{BOOLEXP
:boolop and
:args (
  {OPEXP
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 1
:varattno 6
:varstype 1700
:varstypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
}
{FUNCEXP
:funcid 1740
:funcresulttype 1700
:funcsetset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
  {CONST
:consttype 23
:conststypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
}
:location -1
}
)
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname n
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname n
      :colnames ("n_nationkey" "n_name" "n_regionkey" "n_
comment")
    }
  }
)
}

```

```

:rtkind 0
:relid 18130
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
}
:rtkind 0
:relid 18136
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18130
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18136
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
)
:location -1
}
)
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
)
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0

```

```

        :inputcollid 100
        :args (
          {VAR
            :varno 2
            :varattno 2
            :vartype 1042
            :vartypmod 29
            :varcollid 100
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 2
            :varattnosyn 2
            :location -1
          }
          {CONST
            :consttype 1042
            :consttypmod -1
            :constcollid 100
            :constlen -1
            :constbyval false
            :constisnull false
            :location -1
            :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
          }
        )
        :location -1
      }
    }
    :location -1
  }
  :mergeActionList <>
  :mergeTargetRelation 0
  :mergeJoinCondition <>
  :targetList (
    {TARGETENTRY
      :expr
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
      }
      :resno 1
      :resname ?column?
      :ressortgroupref 0
      :resorigtbl 0
      :resorigcol 0
      :resjunk false
    }
  )
  :override 0
  :onConflict <>
  :returningOldAlias <>
  :returningNewAlias <>
  :returningList <>
  :groupClause <>
  :groupDistinct false
  :groupingSets <>
  :havingQual <>
  :windowClause <>
  :distinctClause <>
  :sortClause <>
  :limitOffset <>
  :limitCount <>
  :limitOption 0
  :rowMarks <>
  :setOperations <>
  :constraintDeps <>
  :withCheckOptions <>
  :stmt_location -1
  :stmt_len -1
}
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname s_suppkey
    :ressortgroupref 0
    :resorigtbl 18153
    :resorigcol 1
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 2
      :vartype 1042
      :vartypmod 29
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location -1
    }
    :resno 2
    :resname s_name
    :ressortgroupref 0
    :resorigtbl 18153
    :resorigcol 2
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 3
      :vartype 1043
      :vartypmod 44
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 3
      :location -1
    }
    :resno 3

```

```

:resname s_address
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname s_nationkey
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname s_phone
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname s_acctbal
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1043
:vartypmod 105
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname s_comment
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 7
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier true
:relid 18621
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 18130
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS

```

```

:aliasname ANY_subquery
:colnames <>
}
:eref
{
  ALIAS
  :aliasname ANY_subquery
  :colnames ("ps_supkey")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{
  ALIAS
  :aliasname partsupp_secure
  :colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtekind 1
:subquery
{
  QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs false
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSubLinks true
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity false
  :hasGroupRTE false
  :isReturn false
  :cteList <>
  :rtable (
    {RANGETBENTRY
      :alias
      {
        ALIAS
        :aliasname ps
        :colnames <>
      }
      :eref
      {
        ALIAS
        :aliasname ps
        :colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplyco
st" "ps_comment")
      }
      :rtekind 0
      :relid 18163
      :inh true
      :relkind r
      :rellockmode 1
      :perminfoindex 1
      :tablesample <>
      :lateral false
      :inFromCl true
      :securityQuals <>
    }
  )
  :rteperminfos (
    {RTEPERMISSIONINFO
      :relid 18163
      :inh true
      :requiredPerms 2
      :checkAsUser 10
      :selectedCols (b 8 9 10 11 12)
      :insertedCols (b)
      :updatedCols (b)
    }
  )
}
:jointree
{
  FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
}
:quals
{
  BOOLEXP
  :boolop and
  :args (
    {OPEXP
      :opno 521
      :opfuncid 147
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 3
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 3
          :location -1
        }
        {CONST
          :consttype 23
          :consttypmod -1
          :constcollid 0
          :constlen 4
          :constbyval true
          :constisnull false
          :location -1
          :constvalue 4 [ 100 0 0 0 0 0 0 ]
        }
      )
    }
    :location -1
  )
  {SUBLINK
    :subLinkType 0
    :subLinkId 0
    :testexpr <>
    :operName <>
    :subselect
    {
      QUERY
      :commandType 1
      :querySource 0
      :canSetTag true
      :utilityStmt <>
      :resultRelation 0
      :hasAggs false
      :hasWindowFuncs false
      :hasTargetSRFs false
      :hasSubLinks false
      :hasDistinctOn false
      :hasRecursive false
      :hasModifyingCTE false
    }
  )
}

```

```

:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelList <>
:rtable (
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname s
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname s
      :colnames ("s_supkey" "s_name" "s_address" "s_nati
onkey" "s_phone" "s_acctbal" "s_comment")
    }
    :rtekind 0
    :relid 18153
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 18153
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromList (
    {RANGETBREF
      :rtindex 1
    }
  )
  :quals
  {BOOLEXP
    :boolop and
    :args (
      {OEXPR
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
          }
          {VAR
            :varno 1
            :varattno 2
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 1
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 2
            :location -1
          }
        )
      }
      :location -1
    }
  )
  {OEXPR
    :opno 1756
    :opfuncid 1720
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 6
        :vartype 1700
        :vartypmod 983046
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 6
        :location -1
      }
      {FUNCEXP
        :funcid 1740
        :funcresulttype 1700
        :funcset false
        :funcvariadic false
        :funcformat 2
        :funcollid 0
        :inputcollid 0
        :args (
          {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 4 [ 0 0 0 0 0 0 0 ]
          }
        )
      }
      :location -1
    }
  )
  :location -1
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
    }
  }
)

```

```

        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
}
:location -1
}
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
    }
    :resno 1
    :resname ps_partkey
    :ressortgroupref 0
    :resorigtbl 18163
    :resorigcol 1
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
    }
    :resno 2
    :resname ps_supkey
    :ressortgroupref 0
    :resorigtbl 18163
    :resorigcol 2
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
    }
    :resno 3
    :resname ps_avallqty
    :ressortgroupref 0
    :resorigtbl 18163
    :resorigcol 3
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 4
    :vartype 1700
    :vartypmod 983046
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location -1
    }
    :resno 4
    :resname ps_supplycost
    :ressortgroupref 0
    :resorigtbl 18163
    :resorigcol 4
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 5
    :vartype 1043
    :vartypmod 263
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5

```

```

        :location -1
    }
    :resno 5
    :resname ps_comment
    :ressortgroupref 0
    :resorigtbl 18163
    :resorigcol 5
    :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier true
:relid 18625
:inh false
:relkind v
:rellockmode 1
:perminfoindex 3
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTRY
:alias
{ALIAS
:aliasname ANY_subquery
:colnames <>
}
}
:eref
{ALIAS
:aliasname ANY_subquery
:colnames ("p_partkey")
}
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBLENTRY
:alias <>
}
:eref
{ALIAS
:aliasname part
:colnames ("p_partkey" "p_name" "p_mfg" "p_brand" "p_type" "p_size"
"p_container" "p_retailprice" "p_comment")
}
}
:rtekind 0
:relid 18141
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18621
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18130
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18625
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
}
{RTEPERMISSIONINFO
:relid 18141
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
)
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
}
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (

```

```

{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 1254
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 1268
}
}
:location 1266
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location 1292
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 1301
:constvalue 10 [ 40 0 0 0 70 82 65 78 67 69 ]
}
}
:location 1299
}
}
}
:rarg
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 4
}
)
:quals (
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{FUNCEXPR
:funcid 1740
:funcresulttype 1700
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location 657
}
)
:location -1
}
{SUBPLAN
:sublinkType 4
:testexpr <>
:paramIds <>
:plan_id 1
:plan_name SubPlan\ 1
:firstCollType 1700
:firstCollTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse false
:parallel_safe false
:setParam <>
:parParam (i 0 1)
:args (
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 967
}
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4

```

```

        :varattnosyn 2
        :location 1024
    }
}
:startup_cost 0
:per_call_cost 42.26
}
}
:location 669
}
}
}
:rarg
{FROMEXPR
:fromList (
{RANGETBLREF
:rtindex 6
}
)
:quals (
{OPEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{RELABELTYPE
:arg
{VAR
:varno 6
:varattno 2
:vartype 1043
:vartypmod 59
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 2
:location 581
}
}
:resulttype 25
:resulttypmod -1
:resultcollid 100
:relabelformat 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 593
:constvalue 11 [ 44 0 0 0 37 105 118 111 114 121 37 ]
}
}
}
:location 588
}
}
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 316
}
{VAR
:varno 6
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 1
:location 411
}
}
}
:location 327
}
}
}
:alias <>
:rtindex 0
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 136
}
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 198
}
}
}
:location 146
}

```

```

    )
    :alias <>
    :rtindex 0
  }
}
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
  :expr
  {VAR
  :varno 1
  :varattno 2
  :vartype 1042
  :vartypmod 29
  :varcollid 100
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 2
  :location 51
  }
  :resno 1
  :resname s_name
  :ressortgroupref 1
  :resorigtbl 18621
  :resorigcol 2
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 1
  :varattno 3
  :vartype 1043
  :vartypmod 44
  :varcollid 100
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 3
  :location 67
  }
  :resno 2
  :resname s_address
  :ressortgroupref 0
  :resorigtbl 18621
  :resorigcol 3
  :resjunk false
  }
  )
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
  :tleSortGroupRef 1
  :eqop 1054
  :sortop 1058
  :reverse_sort false
  :nulls_first false
  :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
)

```

WARNING: is_parallel_safe: INPUT qual = <>

WARNING: is_parallel_safe: OUTPUT qual = (<VAR :varno 1 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location

-1) {VAR :varno 1 :varattno 3 :vartype 1043 :vartypmod 44 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 1 :varattno 4 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})

WARNING: is_parallel_safe: OUTPUT qual = (<VAR :varno 1 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) {VAR :varno 1 :varattno 3 :vartype 1043 :vartypmod 44 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: Log 2 : After pull up At starting of Path Generation:

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:retlist <>
:rtable (
  {RANGETBLENTY
  :alias
  {ALIAS
  :aliasname s
  :colnames <>
  }
  }
  :eref
  {ALIAS
  :aliasname s
  :colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
  }
  :rtekind 0
  :reliid 18153
  :inh false
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
  }
  {RANGETBLENTY
  :alias
  {ALIAS
  :aliasname n
  :colnames <>
  }
  }
  :eref

```

```

        {ALIAS
        :aliasname n
        :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
        }
:rtekind 0
:relid 18130
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 18136
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18153
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18130
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18136
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
)
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
)
:location -1
}
)
}
)
:rarg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
)
:quals <>
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0

```

```

:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 4
:varatype 23
:varattpmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
)
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 3
:varatype 23
:varattpmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location -1
}
{VAR
:varno 3
:varattno 1
:varatype 23
:varattpmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
)
:location -1
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 3
:varattno 2
:varatype 1042
:varattpmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:consttppmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
)
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varatype 23
:varattpmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
}
:resno 1
:resname s_supkey
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:varatype 1042
:varattpmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
:resno 2
:resname s_name
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 2
:resjunk false
}
{TARGETENTRY

```

```

:expr
{VAR
:varno 1
:varattno 3
:vartype 1043
:vartypmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname s_address
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname s_nationkey
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 19
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 5
:resname s_phone
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1700
:consttypmod 983046
:constcollid 0
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 6
:resname s_acctbal
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1043
:consttypmod 105
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 7
:resname s_comment
:ressortgroupref 0
:resorigtbl 18153
:resorigcol 7
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:smt_location -1
:smt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1756 :opfncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 17
00 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :cons
tbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof fals
e :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_seria
l 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsiz
e -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashegoperator 0 :right_hashegoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1756 :opfncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 1
700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :con
stbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof fal
se :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_seri
al 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsiz
e -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashegoperator 0 :right_hashegoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
{VAR :varno 1 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 1 :varattno 3 :vartype 1
043 :vartypmod 44 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnull
ingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
{VAR :varno 2 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 3 :location -1})

```



```

:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
    :alias
      {ALIAS
        :aliasname ps
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname ps
        :colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "p
          s_comment")
      }
    :rtekind 0
    :relid 18163
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBLENTY
    :alias
      {ALIAS
        :aliasname s
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname s
        :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
          s_acctbal" "s_comment")
      }
    :rtekind 0
    :relid 18153
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rtepermissions (
  {RTEPERMISSIONINFO
    :relid 18163
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 9 10 11 12)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 18153
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {JOINEXPR
        :jointype 4
        :isNatural false
        :larg
          {FROMEXPR
            :fromlist (
              {RANGETBLREF
                :rtindex 1
              }
            )
            :quals (
              {OPEXPR
                :opno 521
                :opfuncid 147
                :opresulttype 16
                :opretset false
                :opcollid 0
                :inputcollid 0
                :args (
                  {VAR
                    :varno 1
                    :varattno 3
                    :vartype 23
                    :vartypmod -1
                    :varcollid 0
                    :varnullingrels (b)
                    :varlevelsup 0
                    :varreturningtype 0
                    :varnosyn 1
                    :varattnosyn 3
                    :location -1
                  }
                  {CONST
                    :consttype 23
                    :consttypmod -1
                    :constcollid 0
                    :constlen 4
                    :constbyval true
                    :constisnull false
                    :location -1
                    :constvalue 4 [ 100 0 0 0 0 0 0 0 ]
                  }
                )
              }
            )
            :location -1
          }
        )
      }
    )
    :rarg
      {RANGETBLREF
        :rtindex 2
      }
    :usingClause <>
    :join_using_alias <>
    :quals (
      {OPEXPR
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR

```

```

:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
:location -1
}
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
)
:location -1
}
}
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname ps_partkey
:ressortgroupref 0
:resorigtbl 18163
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname ps_suppkey
:ressortgroupref 0
:resorigtbl 18163
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname ps_availqty
:ressortgroupref 0
:resorigtbl 18163
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1700
:consttypmod 983046
:constcollid 0
:constlen -1
:constbyval false
:constisnull true

```

```

:location -1
:constvalue <>
}
:resno 4
:resname ps_supplycost
:ressortgroupref 0
:resorigtbl 18163
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:exp:
(CONST
:consttype 1043
:consttypmod 203
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 5
:resname ps_comment
:ressortgroupref 0
:resorigtbl 18163
:resorigcol 5
:resjunk false
}
}
:override 0
:onconflict <>
:repartitioningoldalias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:fromMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
}

WARNING:  is_parallel_safe: INPUT qual = (({RESTRICINFO :clause (OEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) (CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 100 0 0 0 0 0 0 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: OUTPUT qual = (({RESTRICINFO :clause (OEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) (CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 100 0 0 0 0 0 0 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen 1 :constbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b 2) :left_relds (b 2) :right_relds (b 2) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ((VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen 1 :constbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ])) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b 1) :left_relds (b 2) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0.0025 :norm_select -1 :outer_select 0.9085 :mergeofamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOIN :jdomain :j_d_relds (b 1 2))) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOIN :jdomain :j_d_relds (b 1 2))) :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 96 :right_hashoperator 96)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen 1 :constbyval false :constisnull true :location -1 :constvalue <>)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen 1 :constbyval false :constisnull true :location -1 :constvalue <>)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen 1 :constbyval false :constisnull true :location -1 :constvalue <>)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen 1 :constbyval false :constisnull true :location -1 :constvalue <>)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen 1 :constbyval false :constisnull true :location -1 :constvalue <>)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn
```

QUERY PLAN

```
0 loops=1)
```

```

Output: lineitem_secure.l_orderkey, lineitem_secure.l_partkey, lineitem_secure.l_suppkey, lineitem_secure.l_linenum, lineitem_secure.l_quantity, lineitem_secure.l_extendedprice, lineitem_secure.l_discount, lineitem_secure.l_tax, lineitem_secure.l_returnflag, lineitem_secure.l_linestatus, lineitem_secure.l_shipdate, lineitem_secure.l_commitdate, lineitem_secure.l_receiptdate, lineitem_secure.l_shipinstruct, lineitem_secure.l_shipmode, lineitem_secure.l_comment
Filter: (lineitem_secure.l_shipdate < '1996-01-01 00:00:00'::timestamp without time zone)
Rows Removed by Filter: 1
Buffers: shared hit=3382915 read=225755 written=261
-> Nested Loop Semi Join (cost=1.27..42.23 rows=1 width=357) (actual time=0.022..0.043 rows=1 loops=142671)
Output: NULL::integer, NULL::integer, NULL::integer, NULL::integer, 1.l_quantity, NULL::numeric(15,2), NULL::numeric(15,2), NULL::character(1), NULL::character(1), 1.l_shipdate, NULL::date, NULL::date, NULL::character(25), NULL::character(10), NULL::character_varying(44)
Buffers: shared hit=3382915 read=225755 written=261
-> Nested Loop Semi Join (cost=0.85..33.78 rows=1 width=17) (actual time=0.020..0.040 rows=1 loops=142671)
Output: 1.l_quantity, 1.l_shipdate, 1.l_partkey, 1.l_suppkey
Buffers: shared hit=2649455 read=225753 written=261
-> Index Scan using idx_lineitem_partkey_suppkey_shipdate on public.lineitem l (cost=0.43..8.45 rows=1 width=21) (actual time=0.007..0.021 rows=4.28 loops=142671)
Output: 1.l_orderkey, 1.l_partkey, 1.l_suppkey, 1.l_linenum, 1.l_quantity, 1.l_extendedprice, 1.l_discount, 1.l_tax, 1.l_returnflag, 1.l_linestatus, 1.l_shipdate, 1.l_commitdate, 1.l_receiptdate, 1.l_shipinstruct, 1.l_shipmode, 1.l_comment
Index Cond: ((1.l_partkey = partsupp_secure.ps_partkey) AND (1.l_suppkey = partsupp_secure.ps_suppkey) AND (1.l_shipdate >= '1995-01-01'::date))
Index Searches: 142671
Buffers: shared hit=817269 read=224269 written=261
-> Index Only Scan using idx_orders_orderkey_priority on public.orders o (cost=0.42..12.88 rows=1 width=4) (actual time=0.004..0.004 rows=0.40 loops=611223)
Output: o.o_orderkey, o.o_orderpriority
Index Cond: ((o.o_orderkey = 1.l_orderkey) AND (o.o_orderpriority = ANY ('{1-URGENT,2-HIGH}'::bpchar[])))
Heap Fetches: 0
Index Searches: 611223
Buffers: shared hit=1832186 read=1484
-> Index Only Scan using idx_partsupp_partkey_suppkey_availqty on public.partsupp ps_1 (cost=0.42..8.44 rows=1 width=8) (actual time=0.002..0.002 rows=1.00 loops=244487)
Output: ps_1.ps_partkey, ps_1.ps_suppkey, ps_1.ps_availqty
Index Cond: ((ps_1.ps_partkey = partsupp_secure.ps_partkey) AND (ps_1.ps_suppkey = partsupp_secure.ps_suppkey) AND (ps_1.ps_availqty > 0))
Heap Fetches: 0
Index Searches: 244487
Buffers: shared hit=733460 read=2
-> Hash (cost=928.06..928.06 rows=808 width=4) (actual time=6.489..6.489 rows=2082.00 loops=1)
Output: part.p_partkey
Buckets: 4096 (originally 1024) Batches: 1 (originally 1) Memory Usage: 106kB
Buffers: shared hit=762
-> Bitmap Heap Scan on public.part (cost=34.01..928.06 rows=808 width=4) (actual time=0.803..6.011 rows=2082.00 loops=1)
Output: part.p_partkey
Recheck Cond: ((part.p_name)::text ~ 'ivory%')::text
Heap Blocks: exact=752
Buffers: shared hit=762
-> Bitmap Index Scan on idx_part_name_trgm (cost=0.00..33.81 rows=808 width=0) (actual time=0.628..0.628 rows=2082.00 loops=1)
Index Cond: ((part.p_name)::text ~ 'ivory%')::text
Index Searches: 1
Buffers: shared hit=10

Planning:
Buffers: shared hit=434 read=27
Planning Time: 35.886 ms
Execution Time: 6428.896 ms
(139 rows)

tpch-200mb-2=>

```